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Towards Visual Analytics for Explainable AI in Industrial Applications

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Abstract: As the levels of automation and reliance on modern artificial intelligence (AI) approaches increase across multiple industries, the importance of the human-centered perspective becomes more evident. Various actors in such industrial applications, including equipment operators and decision makers, have their needs and preferences that often do not align with the decisions produced by black-box models, potentially leading to mistrust and wasted productivity gain opportunities. In this paper, we examine these issues through the lenses of visual analytics and, more broadly, interactive visualization, and we argue that the methods and techniques from these fields can lead to advances in both academic research and industrial innovations concerning the explainability of AI models. To address the existing gap within and across the research and application fields, we propose a conceptual framework for visual analytics design and evaluation for such scenarios, followed by a preliminary roadmap and call to action for the respective communities.

Keywords: explainable artificial intelligence; XAI; human-centered artificial intelligence; visual analytics; industrial applications; human–automation collaboration; information visualization; data visualization

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1. Introduction

Among the variety of approaches and methods for data analytics, the research on and applications of *artificial intelligence* (AI) have attracted the interest of professional data analysts, researchers and practitioners across multiple disciplines/domains, and the general public since the 1950s, especially during the past decade. Advances in data-driven *machine learning* (ML) and *deep learning* (DL) methods, as well as other subfields of AI, have led to the wide adoption of the respective techniques around the world [1]. One recent example is the public interest in approaches based on large language models, exemplified by ChatGPT [2,3]. However, a number of concerns have also been expressed about the conceptual, technological, socioeconomic and ethical aspects of AI development strategies centered on automation [4], including the issues with large language models [5] and ChatGPT in particular [6], for instance. Local, national and international policymakers have also started introducing various regulations that indirectly or directly affect such AI/ML applications [1], including the "right to be informed" of the EU GDPR and the EU AI Act, leading to further discussions of the risks and appropriate actions for research, development and governance of such applications [7]. Thus, efforts to introduce and apply AI/ML in industrial scenarios [8–11] must consider the respective concerns.

Human–AI collaboration presents several cognitive challenges for the end users who interact with it, notably the effect of trust and transparency on the adoption of AI technology [12,13]. When AI systems provide arguments for their decisions (including explanations), it can reshape the way humans process information and understand the world around them [14,15]. AI addresses the human tendency for *satisficing* when reasoning [16,17], in order to reduce the cognitive load while still providing meaningful insights. This may be beneficial in certain contexts (e.g., see the work of Dragoni et al. [14]), but it can also reinforce misconceptions and lead to inappropriate actions. This is particularly sensitive in safety-critical industrial applications, such as aviation [18] and military [12], where human-centered approaches that retain human operators' authority over AI are advocated. *Explainable AI* (XAI) serves as a critical mechanism for calibrating user trust in AI by providing insights into AI/ML system decision-making processes [12]. While explanations can foster understanding and rational trust in AI systems, they simultaneously risk triggering cognitive biases, potentially leading to over-reliance or misunderstandings in the AI system [19,20]. Moreover, increased transparency often equates presenting more information, creating a challenge in balancing comprehensiveness with cognitive manageability. By understanding these cognitive challenges, we can design better human–AI collaborative systems and training programs that enhance the synergy between human intuition and AI capabilities.

To provide a better understanding of the models' internals as well as their behavior and predictions—*interpretability* and *explainability* [21–23]—several approaches have been proposed by the AI/ML research community and practitioners [24,25], including, for instance, salience maps for internal model computations, proxy/post hoc models, and generated explanations. Nauta et al. provide the following definition for such AI/ML scenarios: "An *explanation* is a presentation of (aspects of) the reasoning, functioning and/or behavior of a machine learning model in human-understandable terms" [26].

The XAI community thus acknowledges the importance of going beyond the exclusive use of "black box" models [22]; however, the calls to empower *humans* involved in the AI/ML model construction or utilization have been even stronger from the researchers with backgrounds in *human–computer interaction* (HCI) [27], *information visualization* (InfoVis) [28,29], and especially *visual analytic* (VA) [30–32] communities (see Figure 1), leading to the notions of human-centered AI [33] and human-centered ML [34,35].

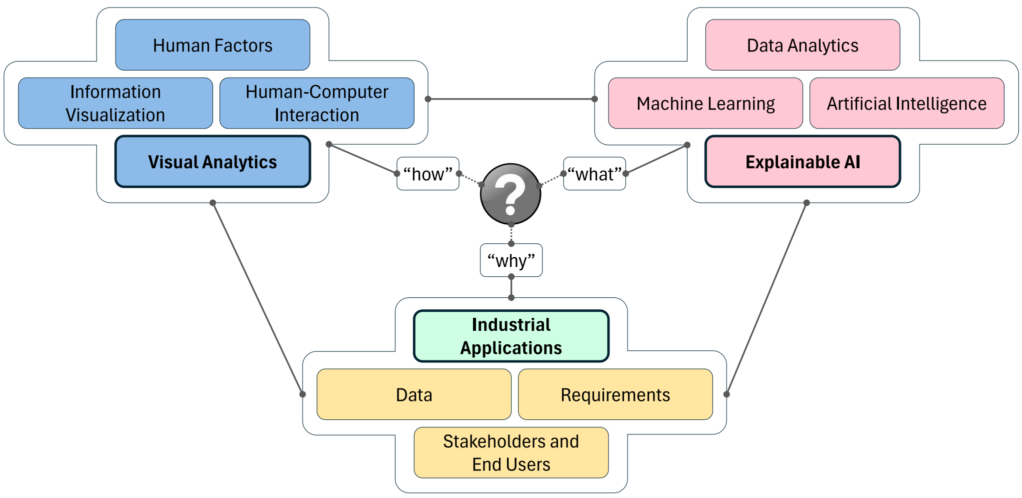

Figure 1. Overview of the disciplines and fields relevant to this study, including the gap identified on the intersection between visual analytics, explainable artificial intelligence and industrial applications with respect to the design and evaluation frameworks and guidelines.

In simple terms, XAI research has primarily concentrated on determining the content of explanations—*what* should be explained. However, an equally crucial aspect is *how* to present these explanations effectively to facilitate understanding. This can be broken down into three interconnected fields: (1) InfoVis methods can be used to decide how to represent raw data, model training results, inference/prediction outputs and explanations visually, from basic line and bar charts to powerful novel metaphors, and also how to interact with such views to support user tasks; (2) VA is concerned with making use of such individual visual representations and interactions alongside computational models (including AI/ML) as part of analytical workflows; while (3) human factors and HCI methods address usability issues and cognitive aspects in interface design and focus on providing accessible and meaningful explanations that consider user needs, mental models and contexts (often with the goal of fostering trust and transparency in AI systems). These fields and the respective methods *can and should be employed* to complement purely computational AI/ML expertise to address the long-term needs for data analysis and the adoption of XAI within industrial applications (the underlying *"why"* context).

However, there currently exists a *gap* within and across the research and application fields (see Figure 1), with a limited number of primary studies and even smaller number of existing guidelines available for utilizing specifically VA methods to support XAI for industrial needs. As summarized in Table 1, the prior secondary and tertiary studies that provided surveys, analyses of open challenges, frameworks or guidelines relevant to these fields typically focused on HCI with no (e.g., see Amershi et al. [36]) or very limited focus on InfoVis/VA (e.g., see Liao et al. [37] or Wright et al. [38]) with respect to XAI in industrial scenarios. The existing work relevant to VA discussed either XAI concerns (e.g., see the publications by Sperrle et al. [39], Alicioglu and Sun [40] or La Rosa et al. [41]) or industrial application concerns (e.g., see Cibulski et al. [42]), but *not* both. The work of Ahmed et al. [11] focuses on XAI in Industry 4.0 and touches upon the importance of HCI and visualization concerns, but does not provide a complete design framework for such applications; and finally, the work of Schreck et al. [43] highlights the role of visualization for achieving visual data science in industrial applications; however, the XAI concerns are only mentioned in passing and no complete design framework is proposed. The focus (and the overall contribution) of this study is thus to make first steps towards identifying and bridging the identified gap (even if it will take a long-term effort from several communities to address it completely); considering Table 1, the goal is thus to address all dimensions (columns).

Table 1. Comparison of the secondary/tertiary study examples (including frameworks, guidelines, etc.) relevant to the scope of our study. A checkmark denotes coverage, while a parenthesized checkmark denotes partial coverage of the respective concerns.

Prior Work	HCI	Visualization	ML/AI	XAI	Industrial Applications
Amershi et al. (2019) [36]	✓		✓		✓
Sperrle et al. (2021) [39]	✓	✓	✓	✓	
Alicioglu and Sun (2022) [40]		✓	✓	✓	
La Rosa et al. (2023) [41]		✓	✓	✓	
Cibulski et al. (2022) [42]	(✓)	✓	(✓)		✓
Liao et al. (2020) [37]	✓	(✓)	✓	✓	✓
Wright et al. (2020) [38]	✓	(✓)	✓	✓	✓
Ahmed et al. (2022) [11]	(✓)	(✓)	✓	✓	✓
Schreck et al. (2023) [43]	(✓)	✓	✓	(✓)	✓

In this paper (motivated by ongoing collaborations), we consider the current state of the art in (1) VA (and, more broadly, interactive visualization) for (2) XAI specifically in (3) industrial applications. We argue that practitioners in the respective domains can benefit from the involvement of visualization techniques and tools when trying to make use of XAI for their needs; at the same time, the visualization community can benefit from collaborations with industry practitioners by identifying and addressing the gaps in interactive visualization support for the respective scenarios, including visual representations, interaction techniques, VA workflows or evaluation/validation methods. The main contributions of this study include the following:

- A review of the prior work relevant to VA for XAI in industrial applications;
- A conceptual framework for VA design and evaluation for such scenarios;
- Analyses of several exemplary instantiations of the proposed framework for prior works;
- A preliminary roadmap for realizing the proposed vision, including a discussion of open challenges to be addressed.

The rest of this manuscript is organized as follows: we first review prior work in several disciplines and fields. In Section 3, we propose and provide an initial validation for a conceptual framework for the problem at hand, followed by a discussion on using the framework as a preliminary roadmap for realizing this vision (as well as limitations of our study) in Section 4 and conclusions in Section 5.

2. Related Work

As this study focuses on interdisciplinary collaborations involving visualization + HCI, (X)AI, and industrial applications, this section discusses prior work relevant to at least partial combinations of efforts from these disciplines and fields.

2.1. Visualization + XAI

The InfoVis research community has produced a large number of techniques and guidelines over the years, which have become especially valuable for amplifying human cognition in exploratory scenarios [29], but also for communication/storytelling purposes [44,45]. Furthermore, the VA community has focused on the problem of supporting various analytical workflows to enable decision making [46], typically relying on computational models including AI/ML methods [31]. Such models and methods could facilitate the data (pre-)processing stages within the interactive visual data exploration workflow—but visualization and interaction could also improve the computational models instead.

This latter possibility, often referred to as "Vis4ML" in the context of ML methods in particular, has been actively pursued lately in relation to human-centered ML [34,35,47]. The aspects and concerns that the VA community has focused on include interactive data labeling [48], enhancing user trust [19,49] in ML with the use of visualization [50], support and applications of data embedding methods [51], communication of predictive models' performance [52], ensuring ML explanations are formed based on human-relevant concepts rather than machine-specific structures [35], VA for XAI [40] and explainable DL specifically [41] as well as (human-centered) evaluation/validation concerns for such methods [39,53].

The focus of the existing research has often been on specific algorithms and models, but also model-agnostic black-box solutions related to feature importance and counterfactual reasoning, including interactive VA approaches such as AdViCE [54], DECE [55] or CoFFI [56]. Such solutions are typically targeted at AI/ML model developers, relying on multiple coordinated views [57] and often powerful, but rather involved workflows. Problems and models that focus on special data types and tasks such as temporal data also result in more specialized solutions [58], such as the VA solution proposed by Schlegel et al. [59] to address both global and local explanation perspectives for time series prediction/forecasting tasks.

Further related work—which does not necessarily focus on novel visual representations or interactions with such views—can be found among HCI contributions, especially the research focusing on user interface design for interactive ML systems [60] and XAI approaches [37], but also human-centered concerns related to data collection and curation [61], for instance.

2.2. Visualization + Industrial Applications

Industrial researchers and practitioners affiliated with data-intensive tasks have long been aware of the power of interactive tools for data wrangling and analysis: as Fisher et al. [62] mentioned in their study of Big Data analytics with 16 professional analysts, one frequent comment was *"Visualization is huge"*. However, business analytics tools often provide relatively limited capabilities in terms of visual representation and interaction, which could be attributed to existing workflows and the desire for simpler, familiar techniques, as indicated by the studies conducted by Kandogan et al. [63] or Russel [64] in business analytics settings. Batch and Elmqvist [65] discuss how data scientists often make use of static visual representations for communication/dissemination at the end of their workflows, while there exists a "visualization gap" (especially with respect to interactive visualization) in the early workflow stages. According to the study by Alspaugh et al. with 30 experienced data analytics [66] on the *exploration* practices, the absolute majority of participants used visualization—in many cases, static rather than interactive representations, but also VA and business intelligence tools (where static representations might be generated and updated based on the interactive UI controls and events, albeit the visual representations themselves are non-interactive). Bartram et al. discuss how rich data tables are used, even when dedicated interactions are not explicitly supported by the respective tools [67].

Still, to successfully support users' analytical reasoning for nontrivial tasks and large and complex datasets, VA systems should facilitate the data analysis process both reactively and proactively. Such solutions should provide various visualization techniques that could help users analyze and interpret the data more effectively [68,69]. When designing and then eventually evaluating visualization systems designed for industrial applications, there are several challenges for VA researchers and practitioners. These challenges include selecting an evaluation method that is appropriate for the design study methodology acquired [70]; recruiting participants who are representative of the target industry [70]; collecting and managing high-quality data that are relevant to industry needs [70,71]; adjusting the design and implementation activities for visual data science techniques to agile development environments prevalent in the industry [43]; managing the complexity of VA tools [71]; understanding the organizational context where data analysis occurs [71]; and ensuring that the interpreted results are communicated to stakeholders and are actionable [72]. The recent study by Walchshofer et al. [73] discusses the relevant difficulties of introducing commercial dashboarding/business intelligence systems (in particular, Power BI) in existing organizations.

To address these challenges, various solutions are proposed. Sedlmair et al. recommend that researchers conduct a variety of evaluation methods and have stakeholders involved throughout the evaluation process [70]. Jänicke et al. recommend participatory design, potentially starting with early prototyping to inform the requirement engineering step [74] which affects the later design, implementation and evaluation stages. The recent study by Cibulski et al. reflected on successful visualization research projects in collaboration with manufacturing industries, highlighting the need to adapt domain practices, technical environments and user onboarding, among others [42].

Several interesting examples of the use of visualization in business/industrial settings include FlowOpt by Barták et al. [75], an interactive tool to support enterprise performance optimization, and OrderMonitor by Tang et al. [76], a VA system for supporting warehouse order processing. Regarding further existing visualization approaches employed in the industry, in addition to the study by Cibulski et al. [42] mentioned above, we refer the readers to the survey by Zhou et al. [77] that also includes scientific 3D visualization and virtual/augmented/mixed reality methods in addition to InfoVis techniques in smart manufacturing scenarios.

2.3. Visualization + XAI + Industrial Applications

The advances in AI/ML over the past decade have naturally affected, or at least triggered, heightened interest from the industrial sector. According to Peres et al., the specific concerns and challenges of industrial AI include industrial-grade infrastructures, Big Data, specialized algorithms, high-quality demands (to support decision-making) and value-oriented objectives [9]. They also raise concerns related to various levels of automation, echoing the concerns and guidelines made from the perspectives of human–automation collaboration [12,78,79] and visualization [80] perspectives. Gröger provides further insights on the data challenges for AI in industrial applications, while acknowledging the role of various user roles and tools, e.g., for data discovery and exploration as part of the data engineering process [10], while De Silva et al. describe a reference architecture for AI-based intelligent industrial informatics, where visualization plays an important role [8].

Ahmed et al. [11] focus specifically on XAI for Industry 4.0, analyzing the applicability of various XAI methods and techniques, e.g., model-specific vs model-agnostic or local vs global explanations, for such scenarios. They highlight the importance of visualization approaches to support XAI in industrial applications with non-restrictive analyses, complementing the algorithmic explanation approaches.

Designs of visualization mechanisms for explanations are essential for ML models, especially in industrial applications where operators' decision-making can have significant consequences on the process they control. As Kotriwala et al. [81] argue, for process industries, for instance, a one-size-fits-all XAI solution is not sufficient considering the particular constraints and experiences of various users (and dynamic or interactive explanation techniques can thus be valuable). It is therefore important to eventually move away from conventional understandability strategies (e.g., quasi-explanations) and towards visually interpretable domain-specific explanations [82]. Limitations and concerns about relying on black-box models have been voiced from a more general perspective [22], but also with respect to studies of human factors in model interpretability within industrial settings [83], including the identified needs to communicate and summarize model behavior, for instance. The recent study by Collaris et al. [84] identified the properties that industrial data scientists expect of local feature importance methods, which are also valuable for the design of appropriate solutions involving visualization.

The existing (and currently limited) body of work on visualization for XAI in industrial applications typically relies on a combination of rather standard visual representations such as line plots, bar charts and heatmaps for the existing (standard or novel) algorithmic explanation techniques such as feature attribution or counterfactual examples, for instance. For instance, Assaf et al. [85] use (1) heatmaps to represent temporal multivariate data, feature attention and time attention, as well as (2) bar charts to represent feature importance for temporal predictions with convolutional neural networks. The application scenarios they describe include photovoltaic plant power production and the prediction of server outage [85]. Manca et al. [86] propose the design of XAIProcessLens, a counterfactual-based dashboard to support XAI in process industries with several levels of explanation abstractions and interactive visual representations of temporal trends and confusion matrices, among others. The What-If Tool by Wexler et al. [87] is designed to facilitate probing and the titular *what-if* analyses with multiple hypotheses by ML practitioners through multiple coordinated views, including several regression scenarios in industrial settings (unfortunately, the authors do not describe specific details about the models and domain problems, probably due to confidentiality concerns). Ciorna et al. [88] describe Interact, another visual *what-if* analysis tool for virtual product design in the tire industry, focusing specifically on ML-based regression methods and analyses of interactions between features through multiple coordinated views.

Several further examples relevant to the use of visualization and XAI methods in industrial applications are also discussed in detail below, but it is clear that there is significant room for further work that would consider these several perspectives jointly already at the initial design stage—and in order to facilitate such a design process, we propose our conceptual framework next.

3. Conceptual Framework for VA for XAI in Industrial Applications

In this section, we propose a conceptual framework for design and evaluation for our topic, followed by exemplary instantiations of the framework for several relevant prior works.

3.1. Conceptual Framework

Figure 2 presents the conceptual framework that could be used to describe existing efforts and to plan and follow up new initiatives in the design and evaluation of (visual) analytic workflows and solutions for (X)AI in industrial applications. This framework is inspired by previous work on visualization and VA design and evaluation, including the nested model by Munzner [89], design study methodology by Sedlmair et al. [70], the data–users–tasks design triangle by Miksch and Aigner [90], the knowledge generation model by Sacha et al. [32], the workflow to improve trust in ML models with the use of visualization [50] and the interdisciplinary workflow for visual text analytics design and evaluation [91]. However, in contrast to these prior models that addressed general concerns or other domain applications, the proposed framework focuses specifically on the support for (X)AI in industrial application scenarios from the perspective of VA. Thus, the concerns related to both computational and visual interactive analyses are addressed in the context of unique circumstances and constraints associated with such scenarios. For example, in addition to concerns related to the design, implementation and evaluation of novel techniques and tools in a purely academic environment, the groups of activities identified in our proposed framework (see the colored boxes at the bottom of Figure 2) can be considered from the perspective of technology readiness level (TRL) [9] perspective, which can be an additional factor for stakeholders to decide whether a certain research collaboration has the potential to produce innovative and attractive outcomes. Reflections on successful visualization research projects in collaboration with manufacturing industries by Cibulski et al. [42] are also taken into account (although the respective study does not focus on XAI methods and tasks). Furthermore, previous contributions made in relation to human factors [19,92] human-centered design [93–96] and human–automation collaboration [12,78,97] have also informed the design of this proposed framework.

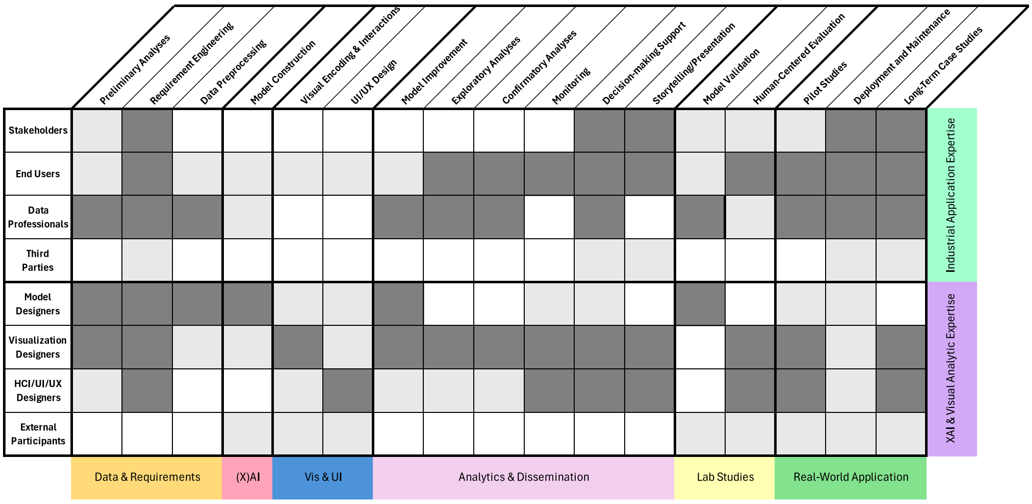

Figure 2. The proposed conceptual framework for visual analytics design and evaluation for XAI in industrial applications. The cell colors within the matrix indicate the assumed involvement levels of various actors in the respective stages and activities for a full-fledged and well-integrated industrial solution (no fill, light grey, and dark grey for no, partial/potential or full involvement, respectively). The overall workflow direction is from left to right; however, multiple implicit feedback loops are expected and should be implemented at most or even all stages in order to ensure agile development and proper project risk management.

In the remainder of this subsection, we discuss the structure of the proposed framework. As indicated in the caption of Figure 2, the overall workflow direction is from left (initial project/collaboration steps with preliminary analyses, etc.) to right (the eventual real-world application concerns including the deployment and long-term use of VA solutions for XAI). At the same time, this overall direction should not be interpreted as a prescription for waterfall-like process organization—most or even all stages of this workflow actually assume interactions between the actors involved as well as feedback loops [70,89]. The colors used in the matrix cells indicate the involvement (full, partial/potential or the lack thereof) of project collaborators in various activities; however, to be used as a design tool, this matrix would be cleared and eventually filled from scratch as a planning tool (possibly even with additions of further rows and columns).

The rows of the matrix in Figure 2 represent *actors* involved in our framework who can be assigned to two groups according to their expertise (potentially, with several roles for the same actor even across the groups):

- **Industrial Application Expertise** This group represents the industrial collaborators who define the domain problem and expect to gain some value [28] from the use of (visual) analytic solutions as project results. The specific *roles* in this group include the following:
  - **Stakeholders** The representatives from the industrial organizations who initiate and/or approve the project collaboration; while they may provide critically important input (such as functional and non-functional requirements) and may test, be informed about or even engage in using the resulting (visual) analytic solutions for XAI, they are not expected to be the main target audience of such solutions. This role could be compared with the *gatekeeper* role described by Sedlmair et al. with respect to the visualization design study methodology [70].
  - **End Users** In contrast to the previous role, end users represent the actual target audience of XAI + (visual) analytic solutions within the respective organizations; for example, process operators monitoring ongoing industrial processes with the help of specialized user interfaces [98]. For successful adoption and trust

in XAI solutions, it is crucial to involve key stakeholders, especially end users and data scientists, from the early stages of development [92]. This proactive involvement can ensure that explanations are tailored to satisfy specific stakeholder needs to effectively bridge understanding gaps and foster trust in the AI system [98,99]. In time-critical environments, designing effective VA for end users poses a significant challenge in that the time available to consult, explore and understand system explanations can be severely constrained by ongoing processes, especially during off-nominal events or anomalies that may require immediate intervention. For such applications, the temporal aspects of presenting VA solutions must be carefully considered to ensure that critical information can be quickly understood and acted upon within the limited time frames of seconds to minutes often available in such high-pressure scenarios.

  - **Data Professionals** This role is played by organization-affiliated professionals who deal with the tasks of data engineering, data science, etc. We expect data professionals to be closely involved in the process of providing the data, participating in data analysis and eventually taking over the long-term deployment and maintenance of XAI + (visual) analytic solutions in the respective organizations. Depending on the specific focus of the project and the desired solution, this role could overlap to a large extent with the previous role of the end user, cf. the *front-line analyst* role discussed by Sedlmair et al. [70].
  - **Third Parties** Finally, we can envision the potential involvement of further parties from the industrial side, such as other collaborating organizations, clients, regulators, researchers or public sector representatives. The respective actors are not directly involved with the design or use of the solutions developed within the scope of collaborations but might be exposed or affected by the respective results, or may also indirectly affect the requirements (e.g., considering the legal or technical regulations).

- **XAI and VA Expertise** This group represents the designers of computational and visual analytic methods to be developed and tested within the scope of project collaboration, either from the same industrial organization or elsewhere (including academia or research institutes/departments):
  - **Model Designers** This role is played by experts in computational analysis, including (X)AI, who are actually responsible for designing and implementing the respective models. While this role might be fulfilled by data professionals from the same organization (as mentioned above), in the general case, it could also be played by academic researchers in (X)AI or specialized X(AI) businesses, for instance.
  - **Visualization Designers** The experts in InfoVis and/or VA who focus specifically on (interactive) visualization concerns to address user tasks in the context of overall solutions. These experts may be internal to the organization or external, such as those from research institutions, academic organizations or specialized businesses.
  - **HCI/UI/UX Designers** Human factors, HCI and UI/UX experts are crucial in shaping the user interface and interaction design for XAI solutions. Kirby and Meyer [100] discuss how this role might be conflated with the previous one in some scenarios or could be fulfilled by dedicated specialists. These specialists focus on creating interfaces that address usability issues and cognitive challenges, ensuring a human-centered approach to XAI. Interfaces should support user sense-making processes, be sensitive to user expertise, context and time constraints, and balance the need for transparency with cognitive load management, especially in high-stake environments [22]. These experts are also essential for

facilitating designs that support trust calibration by helping users understand the AI system's capabilities and limitations.

- **External Participants** Similar to third parties on the industrial application side (as mentioned above), we can envision the involvement of further participants in relation to data, computation and interactive visual analysis design and evaluation stages, such as external data annotators (recruited via crowdsourcing, for instance), laboratory experiment participants (e.g., students, volunteers or Amazon Mechanical Turk workers involved in laboratory experiments for a novel visual representation or UI design feature) or external experts involved in a novel VA method evaluation.

The columns of the matrix in Figure 2 represent *activities and tasks* in our framework that can be grouped in several stages with implicit loops:

- **Data and Requirements** This group comprises the concerns that must be addressed at the initial stages of project collaborations.
  - **Preliminary Analyses** This step might be optional for collaborations that involve participants already familiar with the domain problem and data at hand; however, in many cases initial exploration of the data may be beneficial. The reasons for that may include, for instance, addressing the initial "visualization gap" [65], gaining overview of the data distributions and quality characteristics (e.g., using tools such as OpenRefine, which might later be used for data wrangling), obtaining a better understanding of the domain problem for the rest of the collaborators [100,101], and discussing these initial findings with domain experts in order to address the potential *knowledge gap* [28] and find the *common ground* [91,102]/*common understanding* [100] across (X)AI model developers, visualization and UI designers and industrial domain experts. Jänicke et al. even suggest considering an early functional prototype at this stage to engage the collaborators and make them reflect on their practices and needs before the next steps [74]. From the perspective of enterprise analysts/data scientists, this step could be related to the *discovery* phase described by Kandel et al. [71].
  - **Requirement Engineering** At this stage, problem characterization methods from visualization [70,89,103] and user-centered design [92–94] can be employed alongside requirement engineering techniques from data science and software engineering. The requirements for specific XAI properties, such as explanation transparency [104] or completeness [26], should be introduced and discussed at this point. Furthermore, requirements with a specific focus on visualization in industrial applications [42], industrial AI applications [105] and the respective human factors [83,84] are important to consider here. The requirements for explainability of stakeholders, such as end users, often revolve around bridging trust barriers to facilitate the adoption of AI systems and acceptance of their output. For example, operators working with ML predictions for time series forecasting may want to identify the most influential factors or sensors contributing to a prediction, understand the uncertainty associated with the prediction or contextualize the output by examining similar historical cases, potentially through counterfactual exploration [98,99]. By addressing these needs, XAI solutions can build trust, improve decision-making processes and support the seamless integration of AI systems into industrial operations.
  - **Data Pre-Processing** At this stage, data *acquisition* [65] or *wrangling* [71] takes place. From the visualization perspective, the decisions made at this step can affect *data type abstractions* [89] later made as part of the visual encoding design.

- **(X)AI** At this point, the concerns related to computational and visual interactive components design [32,50,91] could be mapped to the respective tasks and actors.
  - **Model Construction** As part of this activity, the computational model (e.g., based on an XAI approach) is either employed off-the-shelf (if possible), adjusted to the problem at hand or built from scratch. When considering typical ML pipelines for this step, concerns related to data labeling, feature engineering, learning method/algorithm and the final model should be taken into account [50]. From the XAI perspective, this step defines *what* should be explained to the eventual end user (e.g., which decisions/predictions made by the model).
- **Vis and UI** In contrast to computational concerns, this group comprises the steps related to visualization and overall user interfaces for the solutions to be implemented. Depending on the problem and computational model(s), while visualization and UI can be used to provide insight into the model details or decisions, these components might themselves be subject to automation-driven control to some extent [80], e.g., via system *guidance*. From the XAI perspective, this group of steps defines *how* the explanations are presented to the end user.
  - **Visual Encoding and Interactions** Similar to the notions by Munzner [89] and Sedlmair et al. [70], this step is concerned with generating candidate designs for addressing *user tasks* [106] via data abstractions, visual encodings and interactions.
  - **UI/UX Design** In contrast to the previous step, this activity is concerned with the overall user interface and experience considerations based on the HCI principles [27,93,94]. Existing work focusing specifically on design practices for XAI user interfaces is very relevant and valuable here [37,83,104,107].
- **Analytics and Dissemination** While the previous group of activities focuses on *how* to support the relevant (visual) analytic user tasks, we dedicate this group to *why*, i.e., the high-level tasks or goals that the users aim to achieve with respect to the domain problem. These purposes of the analytic activities could be considered from a more general VA [31,32,35] or data science [66,71] perspective, but also more specifically with respect to the (X)AI concerns supported by interactive visualization [41,50,51].
  - **Model Improvement** As Sacha et al. discuss [108], improving the model is the main motivation for many advanced VIS4ML solutions. This high-level task in our proposed framework includes the assistance for model developers during training/debugging, but also quality and bias control of the resulting models.
  - **Exploratory Analyses** This high-level task focuses on discovery and eventual hypothesis generation scenarios facilitated by visual representation and interactive exploration of the raw data, model output/results and the explanations produced by the model.
  - **Confirmatory Analyses** Similar but different from the previous task, this one focuses on the directed search for supporting or conflicting evidence across the raw data, model results and explanations for a hypothesis formulated by the user.
  - **Monitoring** The focus of this high-level task is to provide a view on the ongoing process, including the raw data, model outputs, and explanations—if necessary, with additional affordances aimed to alarm the users (e.g., process operators) in potentially time-critical situations (as opposed to the historical data analysis scenarios unrestricted in time).
  - **Decision-making Support** This high-level task stresses the support of the user in their ongoing decision-making process with real-world implications [109], based on the evidence provided by the tool through its computational results and interactive visual techniques. As Dimara and Stasko mention [109], matching

decision-making tasks to the existing low- and high-level VA task taxonomies is not a trivial step (and neither is the subsequent evaluation of this task). Clarifying the scope of decision-making for the given user and context will be especially important to support users in industrial scenarios with their reliance on increasing sets of data sources and types (as well as the respective data complexity and volume), as highlighted by Schreck et al. [43].

  - **Storytelling/Presentation** Finally, the focus of this high-level task is not to support data analysis per se, but rather to export or organize analytical findings in a form suitable for presentation purposes [44] (e.g., a data analyst compiling their report and presenting it to management).

- **Laboratory Studies** This group of activities focuses on evaluation of individual components of the overall solutions that can be carried out in isolated environments, e.g., when comparing the performance of several alternative models or visual representations.
  - **Model Validation** The performance of computational methods such as ML models is traditionally assessed with respect to specific metrics such as precision, recall, etc. [110], for a given task and ground truth data. However, when considering XAI methods, these approaches might not be sufficient and further concerns mentioned below may also need to be taken into account.
  - **Human-Centered Evaluation** In the fields of visualization and HCI, various forms of human/user-centered evaluation (including the approaches involving expert reviewers) are used to assess usability [111], trust, understanding [92] or other characteristics of visual data representations, interaction techniques, UI designs, etc. In the XAI support scenarios, it is important to also consider the specific factors related to the user's interaction with the model and explanations, e.g., the overall "explainability score" or its more detailed aspects such as explanation transparency, usability and simulatability described by Silva et al. [104], causability [112], or, for instance, 12 Co-properties discussed by Nauta et al. [26]. XAI evaluation metrics for specific data types and modalities have also been proposed, such as time series [113] and image data [114]. Specifically for (X)AI and visualization, we guide the readers to the work by Boukhelifa et al. [53] and Sperrle et al. [39].
- **Real-World Application** Compared to typical academic research activities, collaborations with industrial partners ideally do not stop at the laboratory stage, but rather continue to test the proposed solutions in more realistic scenarios [115], followed by eventual deployment and adoption [116].
  - **Pilot Studies** At this stage, the proposed prototype solution is validated with end users in a (semi)realistic scenario and real-world data provided by the industrial partners.
  - **Deployment and Maintenance** A successfully validated solution is deployed for real-world use, with a stable implementation version potentially maintained over a long life cycle. This and the following points also highlight the need to consider both computational and human-centered aspects that may not be evident for a single limited laboratory study or prototype, e.g., the need to consider the model improvement task discussed above in the context of changing data, models or infrastructure, similar to the recently emerged ML operations (MLOPS) field [117].
  - **Long-Term Case Studies** Even if no active breaking changes are considered for the stable version of the solution deployed, observations about its use (potentially including detailed log data, crash reports or usage statistics), evidence of adoption and additional feedback from industrial partners can provide important

insights for research and innovation purposes [118], including generalizability evidence [47] and inform further improvements or new collaborative projects.

The matrix template for instantiating our framework is available via the link listed in the Supplementary Materials section of this manuscript.

3.2. Exemplary Instantiations

In order to demonstrate the utility and provide a preliminary validation of our proposed framework, we now retroactively instantiate it for several prior works focusing on visualization and VA for industrial applications. By analyzing the respective cases in a similar way to Sacha et al. [32], we test the expressive power of our framework, illustrate the differences between existing work focusing on computational and/or visual analyses in industrial applications and highlight the potential improvement and innovation opportunities (e.g., to support VA in order to complement the implemented XAI techniques, or vice versa).

3.2.1. Visual Analytics for Time-Constant Process Industries

Our first example focuses on the work of Zohrevandi et al. [98] that presents the design of an ecological focus+context linked-and-coordinated VA dashboard to support decision-making for a variety of target audience. These include process operator end users in the paper pulp production industry, stakeholders of time-constant process industries, visualization designers, and HCI/UI/UX designers. Taking an ecological interface design approach [93,94], the designed dashboard aims to support operators in developing an accurate and effective mental model of the process through the visualization of constraints and complex relationships between process control parameters.

The dashboard consists of four main views, each supporting different aspects of the process-related tasks. The tasks are derived on the basis of a work domain analysis performed on the process-specific tasks during a week-long field study as follows: (1) Find the most relevant data points in the historical data that the operator should use as target to maintain steady operations. This is achieved by interacting with an ML classifier. (2) Assess the criteria that would keep the process steady for the relevant process parameters. (3) Assess how different strategies impact the stability of the process.

Figure 3 depicts the dashboard introduced in the respective study [98]. The topological view presents the historical data of a digester vessel used for paper pulp production. The circle chart enables the decision-making operator to visualize the accuracy (indicated by color intensity) and weights (indicated by circle radius) of an ML-based algorithm that predicts the profile of process-related parameters. The scatter plot shows the historical data of the key performance parameter (mapped on the y-axis) for different stages of the process (mapped on the x-axis). The yellow point is selected by the operator as a relevant goal point (i.e., historical reference for comparison). The situation view allows the operator to compare the profile of the key performance parameter (shown as a solid back line) with the profile of selected historical data as goal point (shown as a dashed line). The blue line shows the prediction for the key performance parameter. The strategy view visualizes the change ranges of relevant parameters made by other operators in the historical data in the form of a parallel-coordinate plot. The box plots indicate most common ranges, and the overlaid yellow bars represent the change range for the selected goal point. The black box around a parameter indicates the operator's selection to visualize the model's prediction as well as probe what-if strategies for this parameter. The line plots in the control view show the profile of the selected parameter (indicated by the black line) and the model prediction for the selected parameter (indicated by the blue line). The dotted line shows the profile of the parameter selected for the target point. The horizontal bars allow operators to probe control strategies and visually compare the effects on the selected parameter using the line plot (shown as a colored line, which in the shown example is red for the temperature). The operators can also visualize the effect of the probed strategy on the key performance parameter shown on the situation view.

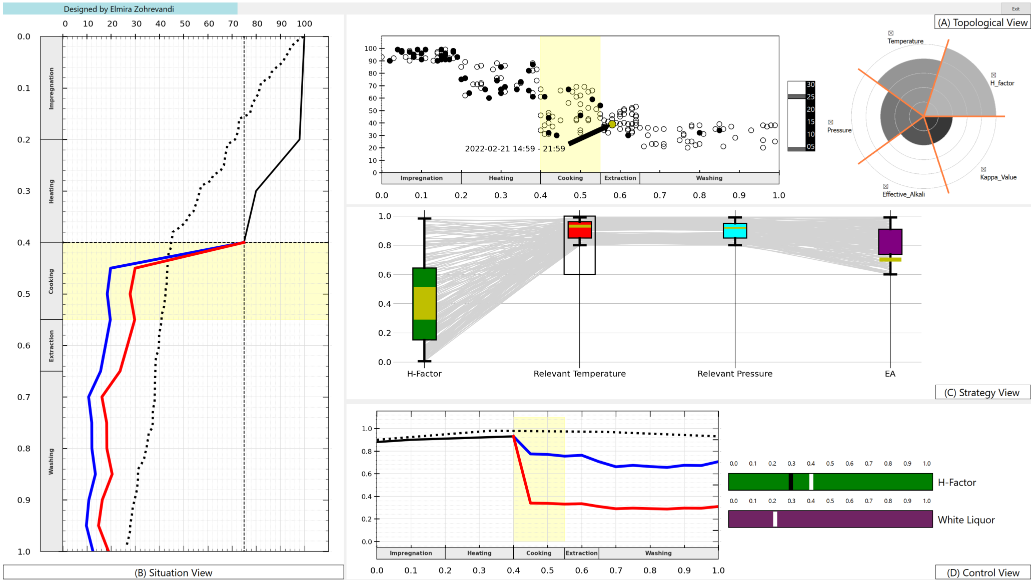

Figure 3. The ecological visual analytics interface presented in Zohrevandi et al. [98], consisting of four views. (**A**) The topological view presents the historical data of how other operators controlled the plant. (**B**) The situation view allows the operator to compare the profile of the key performance parameter with the profile of a selected historical data reference as a goal point. (**C**) The strategy view allows operators to visualize viable change ranges made at previous times (by other operators or oneself) and compare them with goal point data. (**D**) The control view allows operators to probe what-if and what-else strategies and compare their effects on the selected and key performance parameters.

We now provide an analysis of the respective study within the scope of the proposed framework, resulting in the matrix presented in Figure 4. When finding the most relevant data points in the historical data, the tasks *Data and Requirements* are partially supported by its target audience. While the dashboard allows the users to pre-process the historical data that the ML classifier should take into account and also allows them to analyze and compare the historical data points that are found relevant, it does not provide any requirement engineering support.

The dashboard does not support *(X)AI* tasks while it strongly supports *visual encoding and interactions* and *UI/UX Design* tasks for process operator end users, visualization designers and HCI/UI/UX designers.

With respect to *Analytics and Dissemination*, the interface partially supports its target audience. While the model improvement task is not supported at all, exploratory analyses are supported as the interface allows users to navigate through the historical data and explore the most relevant, visualize the viable changes for relevant process parameters and explore the sensitivity of the process to various change ranges for all relevant parameters. Confirmatory analyses are supported by allowing users to compare their selected strategy with the model's prediction of its impact on the process stability. Monitoring is fully supported, as an entire view (called situation view) is dedicated to visualization and prediction of the key process parameter profile. Storytelling/presentation is also supported as stratified views on the dashboard allow users to visualize the connections and links between relevant strategies, control parameters and their impact on process stability. The decision-making task is the main focus of Zohrevandi et al.'s dashboard and is fully supported through the designed *what-if* probe functionality. Following an ecological approach, this functionality supports process operators in weighing in-between control strategies and visually predicting which strategy better keeps the key process parameter steady for a desired time period.

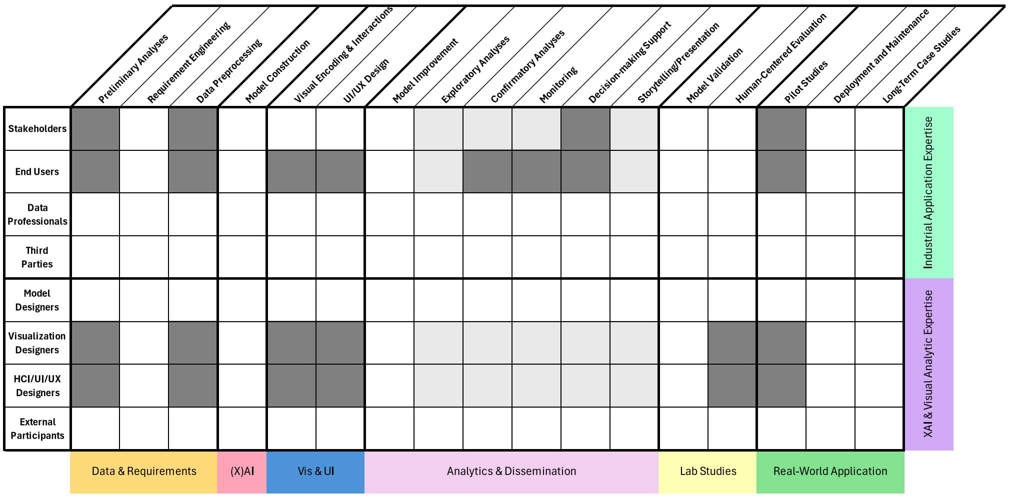

Figure 4. Instantiation of our proposed framework for the visual analytics dashboard for time-constant process industries by Zohrevandi et al. [98]. (no fill, light grey, and dark grey for no, partial/potential or full involvement, respectively).

Laboratory Studies are partially supported for visualization and HCI/UI/UX designers as the dashboard allows for human-centered evaluation studies. This can be achieved through the logging activities that could later be used to evaluate the usefulness of the designed dashboard for its target audience.

Finally, *Real-world Application* is also partially supported by conducting pilot studies. This is provided to stakeholders and process operators who wish to assess their effectiveness on the designed dashboard versus the currently existing tools they use. The pilot studies could also be conducted by visualization designers and HCI/UI/UX designers who wish to extend the ecological approach followed to the design of other dashboards applied to similar industrial applications.

3.2.2. Dynamic Causal Analysis for Industrial Alarm Floods

Manca et al. [86] present a visualization tool (see Figure 5) that aims to support industrial operators in effectively managing the suddenly emerging alarm series by designing an operator-centric comparative causality analysis method. Benefiting from a non-linear causality estimator, the tool incorporates operators' domain-specific knowledge together with process historical data. It supports industrial process operators in managing process-related system alarms through the following. (1) Enabling operators to identify and prioritize critical alarms by visualizing the cause and effect relationships between process variables, as well as disturbance propagation during alarm floods. (2) Time-variant analysis of disturbance propagation or operator interventions. (3) Supporting the operators in identifying the appropriate time-window for investigating historical data.

As depicted in Figure 6, the framework proposed in this work reveals that *Data and Requirements* are partially supported as the tool enables the selection of a specific time window for the analysis of the trends of the process data and automatically performs a data pre-processing step by normalizing the process data based on alarm thresholds and steady-state values. However, the tool does not explicitly focus on the requirement engineering step. *XAI* is not supported as the tool does not allow operators to change the model construction.

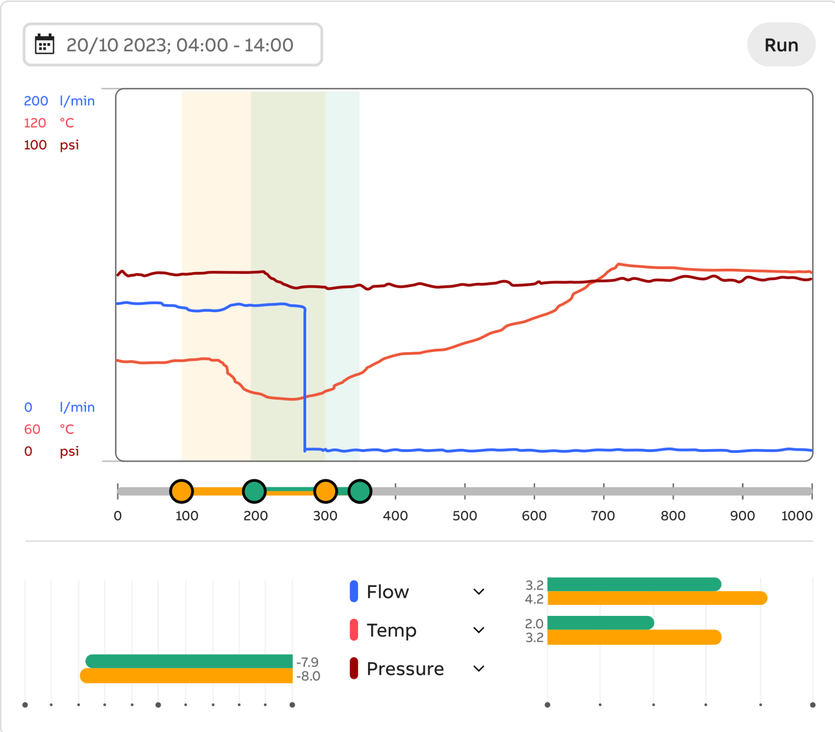

Figure 5. Design of the visual interface for the dynamic causal analysis of industrial alarm floods by Manca et al. [86]. The colored lines depict temporal dynamics of process variables for an operator-specific time window. Below these line graphs, a tornado diagram visualizes causality analysis of the selected process variables. Green and yellow lines enable operators to compare causality analysis outcomes for two distinct time windows. The length of each green and yellow bar represents the range between minimum and maximum causality analysis changes during these two periods. *Image courtesy of Emmanuel Brorsson.*

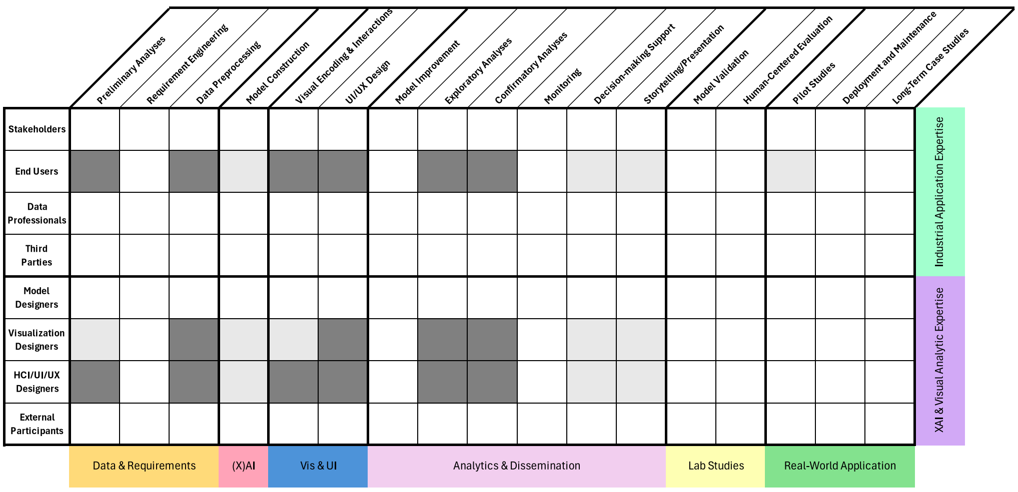

Figure 6. Instantiation of our proposed framework for the dynamic causal analysis of industrial alarm floods by Manca et al. [86]. (no fill, light grey, and dark grey for no, partial/potential or full involvement, respectively).

Vis and UI is strongly supported. The tool uses a tornado diagram to visually represent causal dependencies. In addition, it allows operators to interact with the data and assess various causal relationships. The tool supports exploratory analysis, decision-making and storytelling/presentation, while it does not support model improvements, confirmatory analyses or monitoring. As a result, it partially supports the *Analytics and Dissemination* tasks. Exploratory analyses are supported by visualization of the process behavior and alarm patterns in requested time windows. Decision-making support revolves around the tool's ability to analyze and visualize root causes of alarms, enabling operators to better understand how disturbances propagate, prioritize actions and intervene effectively. Storytelling/presentation is partially supported through the presentation of the results of the causality analysis. However, the tool does not generate a complete narrative as it relies heavily on operators' expertise to interpret the meaning from the visualizations.

This study does not include separate *Laboratory Studies*, as it lacks specific procedures to validate the accuracy or reliability of the causality estimation model. Additionally, it fails to provide information on its usability or effectiveness from an operator's perspective. In contrast, the tool's evaluation uses simulated data, demonstrating its capability to support pilot studies. Thus, it only partially supports the *Real-World Application* tasks. This limitation is due to two factors: first, there is no information on how the tool would be deployed or maintained in an industrial control system environment. Second, the tool does not focus on long-term trends or patterns of alarm floods.

3.2.3. KPI Board

Schmihing et al. [119] describe a VA approach to support quality management in highly flexible production systems. The work employs interactive data visualizations to help production workers analyze data and effectively identify root causes of quality issues. While the main target audience of the designed system is production workers within the automotive sector, their design benefits quality management professionals and production workers of various sectors in the automotive industry whose daily tasks involve the identification and correction of process and data errors.

Analyzing Schmihing et al.'s system, our framework reveals that the designed system fairly supports *Data and Requirements* tasks, lacks support for *XAI* tasks, fairly supports *Vis and UI* tasks, partially supports *Analytics and Dissemination* and *Laboratory Studies* tasks, and poorly supports *Real-World Application* tasks. The outcome of the analysis is shown in Figure 7.

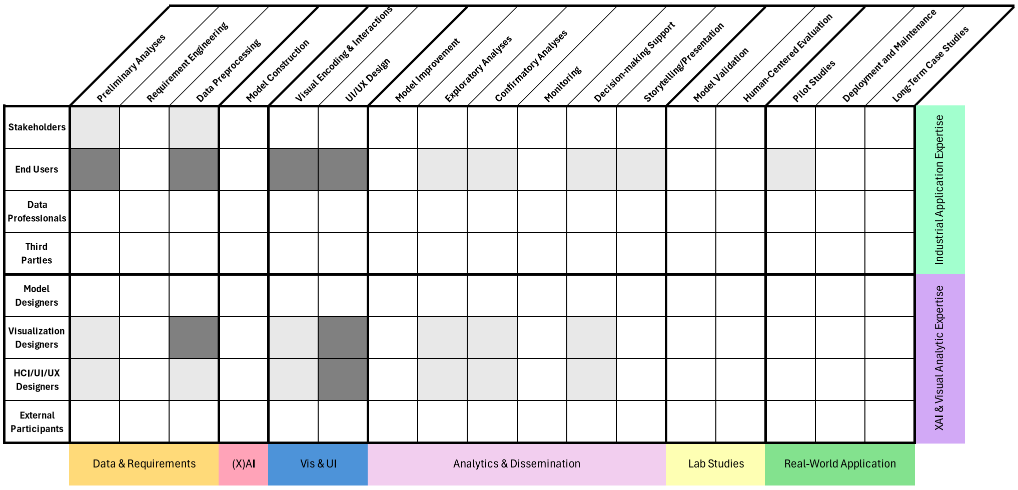

Figure 7. Instantiation of our proposed framework for the KPI Board by Schmihing et al. [119]. (no fill, light grey, and dark grey for no, partial/potential or full involvement, respectively).

Preliminary analyses are supported through enabling users to visualize the distribution of event types and their corresponding required rework time. While requirement engineering is not discussed in detail in this study, the tool automatically conducts data pre-processing by merging the data from different production and quality database sources into a unified data identifier table.

The tool incorporates user feedback to refine its visualization and interaction features. However, it does not support model construction or model improvement, as it lacks any deployed model to predict production quality. However, visual encoding and interaction are strongly supported, and these form the main functionality of the tool. Well-established visualization techniques such as bar charts, scatter plots and timelines are used to enhance the intuitiveness of complex data representations. Visual representations map the relationships between variables and allow users to interact effectively with the data (through filtering, drill-down and selecting data points) to explore these relationships. UI/UX design aspects are also strongly supported, as the design prioritizes user-friendliness and intuitiveness in the dashboard. This is achieved by using visual representations that resemble those currently employed by production workers, fostering a sense of familiarity.

The tool strongly supports exploratory analyses through enabling users to interact with visualizations which helps them identify data trends and anomalies. Filtering and drilling down functionalities into specific data dimensions allow users to generate hypotheses about potential causes of quality issues. The dashboard data are updated in real time (or near real time), which supports the monitoring tasks. This functionality facilitates the early detection of deviations from expected patterns and enhances proactive interventions by workers. Decision-making tasks are partially and indirectly supported. The visualizations allow workers to assess the impact of process changes, but do not explicitly show possible strategies to resolve the issues. Although the tool is not designed for presentation and knowledge sharing purposes, it allows users to capture screenshots or export data summaries to share findings.

Pilot studies and user experiments are indirectly mentioned, but not yet fully conducted (as of the time the respective paper was published). However, future pilot studies and the interface capability in connecting to eye tracking devices are discussed.

3.2.4. XAI Interface for Manufacturing Industry

Grandi et al. [120] propose a methodology to apply XAI concepts to the design of interfaces in manufacturing industries. It particularly focuses on user interface design practices that support operators in the manufacturing industry in gaining insight about complex AI systems. Their methodology follows a matrix-based approach to match different types of explainability visualization with the specific needs of operators during various manufacturing scenarios. The methodology is applied to a case study in agricultural machinery manufacturing to address machine failure challenges in a production plant. The work examines several visualization techniques (text, bar graphs, heatmaps and time series visualizations) to support operators' understanding of crucial information as well as their decision-making. Despite the broader relevance of XAI for a variety of users and stakeholders who need clear explanations of AI-driven systems, the interface designed by Grandi et al. [120] is intended to support operators working in manufacturing environments.

Figure 8 analyzes the interface presented in the work of Grandi et al. [120] using our proposed framework. The framework indicates that the designed interface partially supports *Data and Requirements* tasks, partially supports *XAI* model construction tasks, fairly supports *Vis and UI* tasks, poorly supports *Analytics and Dissemination*, partially supports *Laboratory Studies* tasks and poorly supports *Real-World Application* tasks.

The interface does not support preliminary analyses and data pre-processing tasks. Even though the interface takes the existing systems' data (e.g., the status of a CNC machine), the specifics of how the tool pre-processes the data are not described. However, requirement engineering is strongly supported as a quality function deployment methodology is acquired to match the users' need with appropriate explainability visualization types. This process involves defining scenarios, identifying user requirements through interviews with operators and weighting the requirements based on their importance.

The designed interface partially improves users' understanding and interaction with their designed XAI concepts. This is achieved through their heatmap + barplot visualization, which present a comprehensive overview of anomalies in different sensors, allowing users to compare them using a relationship score.

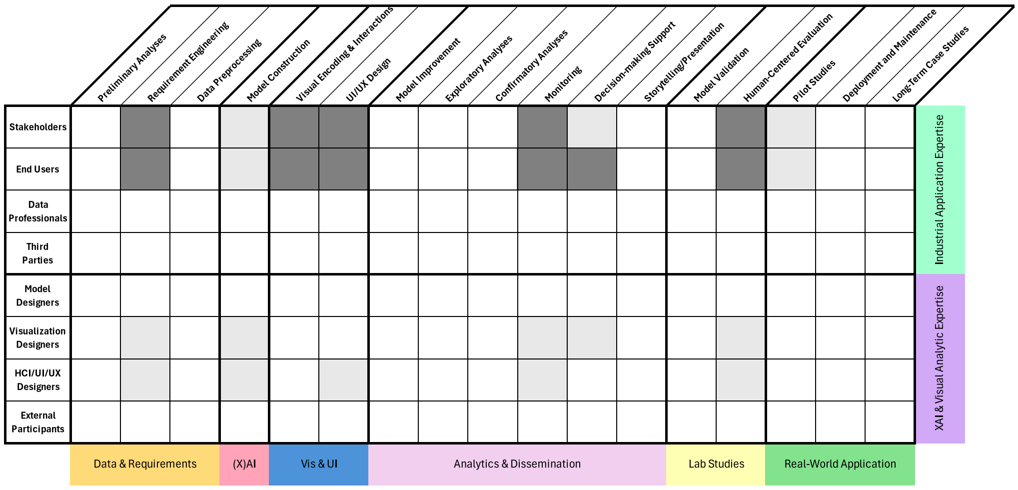

Figure 8. Instantiation of our proposed framework for the methodology presented by Grandi et al. [120] that aims to incorporate XAI concepts into the design of interfaces for manufacturing industry. (no fill, light grey, and dark grey for no, partial/potential or full involvement, respectively).

Visual encoding and interaction is a core functionality of the designed interface. The work uses various explainability visualization techniques to identify and apply the most effective ones for representing the AI model's output to the user. By linking user requirements to specific explainability visualization techniques through a matrix-based methodology, UI/UX design tasks are also supported. The approach provides useful guidelines for creating user interfaces driven by XAI.

The designed interface does not directly involve improving the AI model or supporting exploratory and confirmatory analyses. The monitoring task is, however, supported as in the described case study operators monitor the data of CNC machine status. The interface visualizes time-series plots of sensor data in real time, which enables operators to understand trends leading to potential issues. While storytelling/presentation tasks are not supported at all, the interface partially supports decision-making tasks as it helps operators rank critical features and troubleshoot procedures.

No functionality related to model validation is supported by Grandi et al.'s interface. However, the interface's design and methodology heavily rely on human-centered evaluations. It prioritizes user needs and requirements and matches them with appropriate visualizations on a correlation matrix. Particular use case results reported by the authors from their focus group analysis with four experienced operators indicated the relative importance of suggested visual representations ranging between 3.7% and 20.4%.

The case study conducted with agricultural manufacturing industry as an application of the tool's methodology indicates strong support for pilot study tasks. However, neither the deployment or maintenance of the tool itself nor plans for long-term case studies are discussed by the authors.

4. Discussion

In this section, we focus on the discussion of the next steps to be taken towards the realization of our vision—as well as the limitations of the current study.

4.1. Preliminary Roadmap Towards VA for XAI in Industrial Applications

While the previous section has proposed a framework for the design and evaluation of VA solutions for XAI specifically for industrial applications, we would argue that further steps are required across several communities of academic researchers, industrial researchers and practitioners to make this vision come true. Some of the challenges that should be addressed along this path are discussed next:

- **Motivation for Industrial Application Partners** *"Why bother with interactive visualization or VA at all?"* To address this question, the visualization community should find arguments based on the prior work, e.g., as discussed by van Wijk [28], Fekete et al. [29], Oelke and Sutor [121] or Streeb et al. [122], but also matching the needs experienced by the prospective industrial collaborators (in some cases, due to the current and upcoming AI regulations, for instance) with convincing *success stories* of visualization methods and solutions [42,64,68,69,71,72,74].
- **Motivation for (X)AI Model Designers** The concerns and suggestions above also apply here to some extent, both for practitioners and academic researchers in (X)AI. The visualization community could also refer to the voices from the AI/ML community that recognize the need to focus not only on the algorithm/model, but also data and human in order to improve the overall workflows [10,61]; furthermore, the arguments made for human-centered AI and ML are very much relevant here, such as the works by Shneiderman [33], Sacha et al. [34] and Andrienko et al. [35], among others.
- **Motivation for VA Researchers** The visualization community might also have its reservations for collaborations with a perceived low level of visualization research novelty [28,70,100], considering that the role of applied papers and contributions in the community has been debated over time [123,124]. While the shift in attitude is important, we would like to point out the potential in addressing the present grand challenges in VA applications discussed by Wu et al. [116], including closer intertwining of VA and AI, constructing explainable VA, encouraging deployment, expanding the domain space and user groups, among others. As Wu et al. mention, collaboration with other fields is critically important for the future of the VA field itself—and we would argue that industrial collaborations can complement collaborations with other academic researchers in that regard. The successful examples and identified pitfalls discussed by Cibulski et al. [42] are also valuable and can provide further food for thought for visualization researchers.
- **Common Ground** The prior work accumulated within visualization and HCI can certainly be valuable in this regard [28,74,101,125,126], but even further sources should be considered, e.g., with respect to the challenges of deploying ML solutions [127] or integrating data scientists in software teams [128]. The expectations for industrial AI solutions [8,9,105,129] and specifically interpretability [83] and explainability [11], including human-centered aspects [107,130], should be considered for constructive and successful collaborations.
- **Project Organization and Design Methodology** The framework proposed in this paper—as well as the respective references mentioned in Section 3.1—should provide templates for planning and carrying project collaborations involving several interested parties. The importance of mutual respect and consideration for diverse backgrounds, conventions and needs cannot be overstated here, even if specific assumptions and best practices (e.g., relying on multiple coordinated views [57]) can clash with strong

preferences of end users and even data professionals for quite simplistic visual data representations, interactions and workflows [64,129].

- **Evaluation/Validation** Different disciplines and fields have different traditions and expectations when it comes to verification and validation methods, e.g., the traditional AI/ML approaches may rely on benchmark testing with the available ground truth and performance metrics [110], while the XAI methods may take further considerations regarding the quality of explanations into account [26,104]. The visualization and HCI fields have a rich history of human-centered evaluation methods [115,131], ranging from questionnaires, task-based user studies and eye tracking to assess the usability of specific techniques [111,132–134] to value-driven visualization expert reviews and questionnaires [135,136], usability expert reviews [137], domain expert reviews [138,139] and longitudinal case studies [118]. At the same time, evaluation of complex VA workflows, especially in domain application scenarios, is still considered to be a current grand challenge in the field [116]—and the concerns related to evaluation of VA for XAI bring provide even further nuances [39,53], but also opportunities. In the context of applied collaborations, the expectations from the industrial partners with respect to specific key performance indicators, safety guarantees, scalability [140], etc., should also be considered as part of the validation strategies, same as the generalizability concerns from the VA researchers [47]. Publishing the results of such evaluations based on the common denominator, e.g., an extended version of the template for human-centered evaluations proposed by Sperrle et al. [39], will allow for collection and a common knowledge base valuable for both researchers and practitioners.
- **Replicability vs. Confidentiality** The concerns related to reproducibility, replicability and reliability of (visualization) research results [141] should also be taken into account—and these considerations should be discussed with the industrial partners, who may have their own concerns related to proprietary data, methodology or result confidentiality in mind (e.g., can external experts be invited for a review [139] while proprietary real-world data are being used within the VA solution?), which leads us back to the importance of establishing the common ground and transparent dialogue among the collaborators. Additionally, the common knowledge base concerns mentioned above are also relevant here.
- **Funding and Matchmaking** Finally, some practical considerations besides the scientific and technical topics should be considered to facilitate such collaborations. Both academic and industrial partners should be provided with the necessary resources, but also opportunities for identifying and establishing contacts—while having in mind that a fruitful collaboration typically needs some time and effort to foster. The various actors within the same organizations (e.g., grants and innovation offices at universities), local, national and international facilitators and funding agencies play a critical role in this regard. For example, several funding agencies in Sweden such as Vinnova (https://www.vinnova.se/en/. Last accessed: 1 November 2024) and the Knowledge Foundation (https://www.kks.se/en/. Last accessed: 1 November 2024) focus specifically on the projects with the potential for both scientific and innovative contributions, including various platforms for matchmaking.

4.2. Limitations

Next we discuss several limitations and risks associated with the present study:

- **Framework Reliability** When applying our framework for analysis of the existing tools and workflows, the concerns of reliability similar to inter-/intra-annotator agreement in linguistics or ML data labeling become relevant. Further procedures based on independent coding and computation of agreement measurements such as Cohen's kappa [142] can be helpful for ensuring the annotation reliability in such cases.

- **Iterative vs. Linear Design Workflow** While we have used a matrix-based visual representation to schematically present the steps and actors for our conceptual workflow, we should once again stress the point made in the caption of Figure 2 about the multiple implicit feedback loops. The design process is expected to be iterative with multiple feedback loops and circle back from one step to another, e.g., requirement engineering may follow after each updated version of the prototype, and as it matures from concept, to laboratory tests and eventually real-world testing. The failure to communicate the importance of such feedback loops to the various actors involved in the design process is a potential risk that should be considered when applying our framework; otherwise, the historical reasons for the decline of traditional expert systems may manifest for such inflexible XAI solution designs as well. As Schreck et al. mention with respect to the challenges of designing visual data science solutions for industrial applications, agile development environments rather than long-lasting research planning processes are to be expected in such settings [43].
- **Support for Already Deployed AI Models** Our conceptual framework presented in Section 3.1 implicitly focuses on the more general scenario of designing a solution, system or system of systems of sorts consisting of a *new* ML/AI model, XAI methods (potentially, a separate XAI model such as a surrogate one) and interactive visualization affordances. We should mention that another realistic scenario involves applying XAI and VA methods to an already existing and deployed AI model. While we would argue that our framework is equally applicable to this more special case scenario, the comparison of the effectiveness of the framework across two such settings can be considered part of the future work.
- **Focus on Low TRL Cases** Another limitation of the current study is the focus on interfaces and tools described in research publications, be it purely academic or industrial research and development projects. This opens up opportunities for follow-up studies focusing on analytical applications of our framework for high TRL cases [9], such as commercial products available on the market; yet, another opportunity beyond the scope of this paper would be to apply (and potentially extend, if necessary) our framework as part of a long-term case study, taking the respective project from low to high TRL. Considering the relevant subset of the existing guidelines on human–AI interaction from several major industrial players, as proposed by Amershi et al. [36] or analyzed by Wright et al. [38], could potentially be another valuable addition. Such activities would also help to ensure the external and ecological validity [115] of the proposed conceptual framework.

5. Conclusions

In this study, we propose a conceptual framework to design and evaluate VA interfaces and workflows that support (X)AI solutions in various industrial applications. Our framework provides insights into the strengths and limitations of designed concepts across multiple dimensions, ranging from data handling to real-world implementation. To demonstrate the framework's utility, we applied it to four recent state-of-the-art interfaces, each developed for operators in different process industries: (1) process control in paper pulp production, (2) system alarm flood management, (3) a KPI dashboard for enhancing production quality in the automotive industry and (4) a methodology proposed for incorporating XAI concepts into the design of interfaces for manufacturing companies.

Our framework reveals that the four systems analyzed provide varying levels of support across the task categories. The first interface demonstrates the most comprehensive support with respect to visual encoding and interactions, UI/UX design and decision-making tasks. The second interface shows strong support for visualization and UI tasks and partial support for analytics and dissemination. The third interface excels in data and requirements tasks support. The fourth interface excels in the support of requirement engineering tasks, partially supporting XAI visualization tasks.

Visual encoding interactions and UI/UX design tasks were strongly supported in all four analyzed systems. The emphasis on effective data representation indicates the importance of processes and data complexity management for industrial operators. The analysis also revealed gaps in certain areas, particularly in support for XAI tasks. The lack of explicit support for advanced XAI construction and improvement tasks is a notable weakness in all systems analyzed, which also reflects the very limited number of studies on the combination of visualization and XAI methods for industrial collaborations, which is the subject of our study. However, XAI tasks are increasingly important in industrial applications where AI-driven predictions are commonly used, and understanding the reasoning behind their predictions is crucial for operators.

The reviewed systems vary in their support for analytics and dissemination tasks, with the first and third interfaces particularly notable for their strong support in exploratory analyses and decision-making tasks. The ecological approach of the first system and the emphasis of the third and fourth systems on user-friendliness highlight the importance of considering end-user needs in industrial visualization design.

Although the first, second and third systems support exploratory investigation, they differ in how well they facilitate monitoring, decision-making and confirmatory analyses. Although the first interface excels in supporting decision-making tasks through its novel "what-if" probe, the second system focused on supporting cause-and-effect and root–cause analysis for alarm management. The third system facilitates real-time monitoring capabilities. These demonstrate the variety of VA approaches used in industrial applications. Furthermore, it implies that more comprehensive solutions may result from a more comprehensive design strategy that builds on the advantages of each system, as suggested by our framework.

The analysis revealed a mixed picture regarding laboratory studies and a gap in real-world applications. Although the first, second and fourth systems were validated with pilot studies and human-centered evaluations described in respective publications, there is room for improvement in terms of long-term deployment strategies. This further highlights the need for more rigorous evaluation mechanisms to validate the effectiveness of these visualization systems in actual industrial settings.

Based on our analysis, future developments in designing VA interfaces for industrial applications should highly focus on integrating XAI visual encodings, enhancing real-world application support and conducting more comprehensive evaluations. Moreover, combining the strengths of these systems, i.e., the first system's decision-making support, the second system's causality analysis, the third system's data integration and the fourth system's XAI and requirement engineering approach, may potentially lead to creating more holistic and effective visualization solutions for industrial applications.

To summarize, in this study we have presented our vision for the use of VA for XAI in industrial applications—and while further improvements and validations for the proposed framework and roadmap can be considered part of future work, we would like to conclude this manuscript with a call to action to the respective communities across academia and industry, as we believe there is great potential for scientific advances and innovation resulting from such collaborations.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/analytics4010007/s1. Table S1: matrix template for instantiation of the proposed conceptual framework for visual analytics design and evaluation for XAI in industrial applications.

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